

Milestone 7: Security Implementation and Access Control Configuration

Objective

The objective of this phase was to implement role-based security and access control mechanisms to ensure that users can only access information and functionalities relevant to their responsibilities.

Overview

Security is a critical component of any enterprise application. In the Smart Leave Approval System, role-based access control (RBAC) was implemented to protect leave data and prevent unauthorized access.

The security model ensures that:

- Employees can submit and view their own leave requests.
 - Managers can review and approve leave requests.
 - Administrators have full control over the application.
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Security Architecture

The security architecture was implemented using:

- User Roles
 - Access Control Lists (ACLs)
 - Module Permissions
 - Table-Level Security
 - Field-Level Security
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Role Creation

Employee Role

Role Name

x_1877519_smart_0.employee

Responsibilities

- Apply for leave.
- View personal leave requests.
- Track approval status.

Restrictions

- Cannot approve requests.
 - Cannot modify approval status.
 - Cannot access manager dashboards.
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Manager Role

Role Name

x_1877519_smart_0.leave_manager

Responsibilities

- View leave requests.
 - Approve leave requests.
 - Reject leave requests.
 - Access dashboards and reports.
 - **Restrictions**
 - Cannot modify system configuration.
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Administrator Role

Role Name

admin

Responsibilities

- Full system access.
 - User management.
 - Security management.
 - Dashboard administration.
 - Workflow configuration.
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Access Control Lists (ACLs)

Purpose

ACLs were implemented to control access to records and fields within the Leave Request table.

Table-Level ACLs

Read ACL

Employee Access

Employees can view only their own leave requests.

Manager Access

Managers can view leave requests submitted by employees.

Administrator Access

Administrators can view all records.

Create ACL

Employee Access

Employees can create new leave requests.

Manager Access

Managers can create records if required.

Administrator Access

Administrators have unrestricted access.

Write ACL

Employee Access

Employees can modify requests before approval.

Manager Access

Managers can update approval status.

Administrator Access

Administrators can update all records.

Module Security

Access to modules was restricted using roles.

Apply Leave Module

Assigned Role

x_1877519_smart_0.employee

Purpose

Allows employees to submit leave requests.

My Leave Requests Module

Assigned Role

x_1877519_smart_0.employee

Purpose

Allows employees to view their leave history.

Manager Dashboard Module

Assigned Role

x_1877519_smart_0.leave_manager

Purpose

Provides leave analytics and approval information.

Leave Reports Module

Assigned Role

x_1877519_smart_0.leave_manager

Purpose

Provides reporting capabilities.

Data Protection Measures

The following controls were implemented:

Authentication

Only authenticated users can access the application.

Authorization

Access is controlled through assigned roles.

Data Visibility

Users can only access information relevant to their role.

Record Security

ACLs prevent unauthorized viewing or modification of records.

Security Testing

Test Case 1: Employee Access

Action

Employee logs into the system.

Expected Result

- Can access Apply Leave.
- Can access My Leave Requests.
- Cannot access Manager Dashboard.

Result

Successfully Passed.

Test Case 2: Manager Access

Action

Manager logs into the system.

Expected Result

- Can view leave requests.
- Can approve requests.
- Can access reports and dashboards.

Result

Successfully Passed.

Test Case 3: Unauthorized Access

Action

Employee attempts to access manager functions.

Expected Result

Access denied.

Result

Successfully Passed.

Benefits of Security Implementation

Data Confidentiality

Protects employee leave information.

Controlled Access

Users can only access authorized resources.

Improved Compliance

Supports enterprise security standards.

Reduced Risk

Prevents unauthorized modifications.
